

Grade 8
Unit 13
Lesson 7 Practice

Name Answer Key
Date _____
Period _____

1. How many sides does a regular polygon have if the sum of the interior angles is 3960° ?

$$180(n - 2) = 3,960$$

$$n - 2 = 22$$

$$n = 24 \text{ sides}$$

2. What is the measure of one interior angle of the regular polygon found in the first question? Round to the nearest tenth if necessary.

$$3,960 \div 24 = 165^\circ$$

Use the following for questions 3–5.

The measure of an interior angle in the regular 20 sided figure can be represented by the expression $(19.3x + 7.6)^\circ$.

3. Find the sum of the interior angles.

$$180(20 - 2) = 3,240^\circ$$

4. Find the measure of one interior angle.

$$3,240^\circ \div 20 = 162^\circ$$

5. What is the value of x ?

$$19.3x + 7.6 = 162$$

$$19.3x = 154.4$$

$$x = 8$$

Use the following for questions 6–8.

The measures of two interior angles of a regular polygon are $(14x - 28.2)^\circ$ and $(27 + 10x)^\circ$.

6. Write and solve an equation to find the value of x .

$$14x - 28.2 = 27 + 10x$$

$$4x = 55.2$$

$$x = 13.8$$

7. What is the measure of each interior angle?

$$14(13.8) - 28.2 = 165^\circ$$

8. How many sides does the regular polygon have?

$$180(n - 2) = 165n$$

$$180n - 360 = 165n$$

$$15n = 360$$

$$n = 24 \text{ sides}$$

A bathroom floor consists of tile in the shape of regular octagons and equilateral triangles.

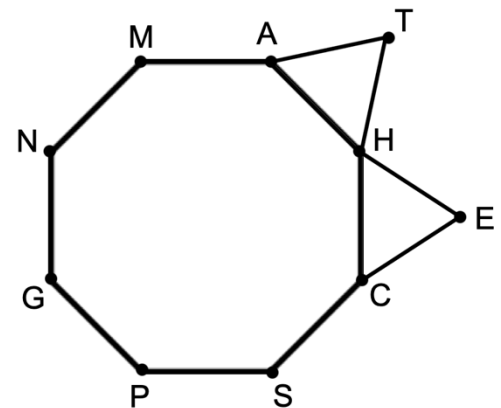
9. Find the sum of $\angle MAH$ and $\angle HAT$.

$$\text{Regular Octagon} = 1,080^\circ$$

$$\angle MAH = 1,080 \div 8 = 135^\circ$$

$$\angle HAT = 180 \div 3 = 60^\circ$$

$$135^\circ + 60^\circ = 195^\circ$$



10. The measure of $\angle MAT$ is $(25x + 15)^\circ$. Determine the value of x .

$$25x + 15 = 360 - (135 + 60)$$

$$25x + 15 = 165$$

$$25x = 150$$

$$x = 6$$